

Here's a more realistic OSPF Single Area lab that includes an ISP router:

OSPF Single Area Lab – Horizon Corp + ISP Connection

Scenario:

Horizon Corp has 3 internal routers. You must configure **OSPF Single Area (Area 0)** for internal routing and advertise a default route to reach the Internet via the ISP.

Topology:

- R1, R2, R3 → Internal routers (OSPF Area 0)
- ISP → External router (represents the Internet)

IP Addressing:

Device	Interface	IP Address	Purpose
R1	G0/0	10.10.10.1/24	LAN1
R1	G0/1	10.10.12.1/30	Link to R2
R1	G0/2	172.16.1.1/30	Link to ISP
R2	G0/0	10.10.20.1/24	LAN2
R2	G0/1	10.10.12.2/30	Link to R1
R2	G0/2	10.10.23.1/30	Link to R3
R3	G0/0	10.10.30.1/24	LAN3
R3	G0/1	10.10.23.2/30	Link to R2
ISP	G0/0	172.16.1.2/30	Link to R1
ISP	Loopback0	8.8.8.8/32	Simulate Internet

PCs:

- PC1: 10.10.10.10/24 (Gateway: 10.10.10.1)
- PC2: 10.10.20.10/24 (Gateway: 10.10.20.1)

- PC3: 10.10.30.10/24 (Gateway: 10.10.30.1)
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Tasks:

1. Configure all interfaces with correct IP addresses.
 2. Configure **OSPF Single Area (Area 0)** on **R1, R2, and R3**.
 3. On **R1**, configure a **default route** pointing to the ISP (172.16.1.2).
 4. Propagate the default route into OSPF so R2 and R3 can reach the Internet.
 5. Verify:
 - All routers form OSPF neighborship.
 - All PCs can ping each other.
 - All PCs can ping the ISP's Loopback (8.8.8.8).
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Bonus Challenges:

- Make R2 the DR on the link between R1 and R2.
 - Configure OSPF authentication between the routers.
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Would you like me to:

- Give you the **step-by-step configuration guide**?
- Provide the expected `show ip route` output?
- Or give you a blank table for VLSM / addressing first?

Just tell me how you want to proceed!